

$$x = b \int \left[\int_{y=f_2(x)}^{y=f_1(x)} f(x, y) dy \right] dx$$

$$x = a \quad \dot{y} = f_1(x)$$

$$\int_{y=c}^{y=d} \left[\int_{x=g_1(y)}^{x=g_2(y)} f(x,y) dx \right] dy$$

$$\int_0^{x^2} \int_{y-x}^y dy dx$$



$y = 0$ $x = y$ $dx \, dy$

$$u = \int \underline{f(x)} dx$$



$$u = a \int_a^b \left[\int_{h_1(x)}^{h_2(x)} f(x, y) dy \right] dx$$

$$\int_a^b \int_{h_1(x)}^{h_2(x)} f(x, y) \cancel{dx} \cancel{dy} \quad \text{or} \quad \int_a^b \int_{h_1(x)}^{h_2(x)} f(x, y) dy dx$$

$$\int_c^d \left[\int_{g_1(y)}^{g_2(y)} f(x, y) dx \right] dy$$

$$u = a \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

$$y = \int_0^1 \int_{x=y}^1 xy e^{-x^2} \underline{dy \, dx} \quad \times$$

$$= \int_0^1 \left[\int_0^y xy e^{-x^2} \underline{dx} \right] dy$$

$$= \int_0^1 y \left(\int_0^y \underline{x e^{-x^2}} \underline{dx} \right) dy$$

$$x^2 = t \Rightarrow 2x \, dx = dt \Rightarrow x \, dx = \frac{dt}{2}$$

$$x: 0 \rightarrow y \quad t: 0 \rightarrow y^2$$

$$= \int_0^1 y \left(\int_0^{y^2} \bar{e}^t \frac{dt}{2} \right) dy$$

$$= \frac{1}{2} \int_0^1 y \left(\frac{\bar{e}^t}{-1} \right)_0^{y^2} dy = \frac{1}{2} \int_0^1 y \left(-\bar{e}^{y^2} - \frac{1}{-1} \right) dy$$

$$= \frac{1}{2} \int_0^1 y (1 - \bar{e}^{y^2}) dy$$

$$y^2 = t \Rightarrow 2y \, dy = dt \quad y \, dy = \frac{dt}{2}$$

as $y: 0 \rightarrow 1 \quad t: 0 \rightarrow 1$

$$= \frac{1}{2} \int_0^1 (1 - \bar{e}^t) \frac{dt}{2} = \frac{1}{4} \left(t + \frac{\bar{e}^t}{-1} \right)_0^1$$

$= \frac{1}{4} (1 + \bar{e}^{-1} - 0 - 1)$

$$\begin{aligned}
& \int_0^1 \int_0^{x^2} e^{(y/x)} \underline{dy} \, dx \\
&= \int_0^1 \left[\frac{e^{y/x}}{1/x} \right]_0^{x^2} dx \\
&= \int_0^1 (x(e^x) - x(1)) \, dx \\
&= \int_0^1 x e^x \, dx - \int_0^1 x \, dx \\
&= \left[x(e^x) - (1)e^x \right]_0^1 - \left(\frac{x^2}{2} \right)_0^1 \\
&= [e - e - 0 + 1 - \frac{1}{2}] = \frac{1}{2}
\end{aligned}$$

$$\int e^{ax} \, dx = \frac{e^{ax}}{a}$$

$$\begin{aligned}
& \int \int u v \\
&= \underline{u \int v} - \int (v u')
\end{aligned}$$

$$\begin{aligned}
& \int \int u v \text{ vanishing der.} \\
&= u \underbrace{\int v}_{v_1} - u' \underbrace{\int v_1}_{v_2} + u' \int v_2
\end{aligned}$$

$$\int_0^\infty \int_0^\infty e^{-x^2(1+y^2)} \underline{\underline{x \, dx \, dy}}$$

$$x^2(1+y^2) = t$$

$$(1+y^2) 2x \, dx = dt$$

$$x \, dx = \frac{dt}{2(1+y^2)}$$

$$x: 0 \rightarrow \infty \quad t: 0 \rightarrow \infty$$

$$\left(\begin{array}{l} e^{-\infty} = \frac{1}{e^{\infty}} \\ = 0 \end{array} \right)$$

$$= \int_0^\infty \int_0^\infty e^{-t} \frac{dt}{2(1+y^2)} dy$$

$$= \frac{1}{2} \int_0^\infty \frac{1}{1+y^2} \left(\int_0^\infty e^{-t} dt \right) dy = \frac{1}{2} \int_0^\infty \frac{1}{1+y^2} \left(\frac{e^{-t}}{-1} \right)_0^\infty dy$$

$$= \frac{1}{2} \int_0^\infty \frac{1}{1+y^2} \left(0 + \frac{1}{1} \right) dy \quad \left| \quad \int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right.$$

$$= \frac{1}{2} \left(\tan^{-1}(y) \right)_0^\infty = \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) = \frac{\pi}{4}$$

$$\int_0^1 \int_{x^2}^x xy(x+y) \, dy \, dx = \int_0^1 \int_{x^2}^x (x^2y + xy^2) \, dy \, dx$$

$$= \int_0^1 \left[x^2 \frac{y^2}{2} + x \frac{y^3}{3} \right]_{x^2}^x dx$$

$$= \int_0^1 \left[\frac{x^4}{2} + \frac{x^4}{3} - \frac{x^2 x^4}{2} - \frac{x x^6}{3} \right] dx$$

$$= \int_0^1 \left[\frac{5x^4}{6} - \frac{x^6}{2} - \frac{x^7}{3} \right] dx =$$

$$= \left[\frac{5}{6} \frac{x^5}{5} - \frac{x^7}{7 \times 2} - \frac{x^8}{3 \times 8} \right]_0^1 = \left[\frac{1}{6} - \frac{1}{14} - \frac{1}{24} \right] = \frac{3}{56}$$

$$\int \sqrt{x^2+a^2} \quad \int \frac{1}{\sqrt{x^2+a^2}}$$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{\underbrace{a^2-y^2}_{\text{}} - x^2} dx dy$$

$$\left[\int \underline{\underline{\sqrt{a^2-x^2}}} dx = \underline{\underline{\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)}} \right]$$

$$= \int_0^a \left(\int_0^{\sqrt{a^2-y^2}} \sqrt{(\sqrt{a^2-y^2})^2 - x^2} \underline{\underline{dx}} \right) dy$$

$$= \int_0^a \left[\frac{x}{2} (\sqrt{a^2-y^2-x^2}) + \frac{(a^2-y^2)}{2} \sin^{-1}\left(\frac{x}{\sqrt{a^2-y^2}}\right) \right]_{\sqrt{a^2-y^2}}^0 dy$$

$$= \int_0^a \left[\frac{\sqrt{a^2-y^2}}{2} (\cancel{\sqrt{a^2-y^2}} - \cancel{(a^2-y^2)}) + \frac{a^2-y^2}{2} \sin^{-1}(1) - 0 \right] dy$$

$$= \int_0^a \left(\frac{a^2-y^2}{2} \right) \frac{\pi}{2} dy = \frac{\pi}{4} \int_0^a (a^2-y^2) dy$$

$$= \frac{\pi}{4} \left[a^2 y - \frac{y^3}{3} \right]_0^a = \frac{\pi}{4} \left[a^3 - \frac{a^3}{3} \right]$$

$$= \frac{\pi}{4} \frac{2a^3}{3} = \underline{\underline{\frac{\pi a^3}{6}}}$$

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dy dx$$

$$= \int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{y^2 + (\sqrt{1+x^2})^2} dy dx \quad \frac{1}{y^2+a}$$

$$= \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} \left(\frac{y}{\sqrt{1+x^2}} \right) \right]_0^{\sqrt{1+x^2}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}} \left[\tan^{-1}(1) - \tan^{-1}(0) \right] dx$$

$$= \int_0^1 \frac{1}{\sqrt{1+x^2}} \left(\frac{\pi}{4} - 0 \right) dx$$

$$= \frac{\pi}{4} \int_0^1 \frac{dx}{\sqrt{x^2+1}} \quad \left. \begin{array}{l} x = \tan \theta \\ dx = \sec^2 \theta d\theta \\ x: 0 \rightarrow 1 \\ \theta: 0 \rightarrow \pi/4 \end{array} \right\}$$

$$= \frac{\pi}{4} \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{\sec \theta} = \frac{\pi}{4} \log(\sec \theta + \tan \theta) \Big|_0^{\pi/4}$$

$$= \frac{\pi}{4} [\log(\sqrt{2}+1) - \log(1)]$$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-y^2-x^2} dx dy$$

$$\int \sqrt{a^2-x^2} = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$$

$$= \int_0^a \left[\frac{x}{2} \sqrt{a^2-y^2-x^2} + \left(\frac{a^2-y^2}{2} \right) \sin^{-1} \left(\frac{x}{\sqrt{a^2-y^2}} \right) \right]_0^{\sqrt{a^2-y^2}} dy$$

$$= \int_0^a \left[\frac{\sqrt{a^2-y^2}}{2} \sqrt{a^2-y^2-(a^2-y^2)} + \left(\frac{a^2-y^2}{2} \right) \sin^{-1}(1) - 0 - 0 \right] dy$$

$$= \int_0^a \left[\frac{\sqrt{a^2-y^2}}{2} \sqrt{a^2-y^2-(a^2-y^2)} + \left(\frac{a^2-y^2}{2} \right) \sin^{-1}(1) - 0 - 0 \right] dy$$

$$= \int_0^a \left(\frac{a^2-y^2}{2} \right) \frac{\pi}{2} dy = \frac{\pi}{4} \int_0^a (a^2-y^2) dy$$

$$= \frac{\pi}{4} \left(a^2 y - \frac{y^3}{3} \right) \Big|_0^a$$

$$= \frac{\pi}{4} \left(a^3 - \frac{a^3}{3} \right) = \frac{\pi a^3}{6}$$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{\underbrace{a^2-y^2}_{\text{}} - x^2} dx dy$$

$$\left[\int \underline{\underline{\sqrt{a^2-x^2}}} dx = \underline{\underline{\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)}} \right]$$

$$= \int_0^a \left(\int_0^{\sqrt{a^2-y^2}} \sqrt{(\sqrt{a^2-y^2})^2 - x^2} \underline{\underline{dx}} \right) dy$$

$$= \int_0^a \left[\frac{x}{2} (\sqrt{a^2-y^2-x^2}) + \frac{(a^2-y^2)}{2} \sin^{-1}\left(\frac{x}{\sqrt{a^2-y^2}}\right) \right]_{\sqrt{a^2-y^2}}^0 dy$$

$$= \int_0^a \left[\frac{\sqrt{a^2-y^2}}{2} (\cancel{\sqrt{a^2-y^2}} - \cancel{(a^2-y^2)}) + \frac{a^2-y^2}{2} \sin^{-1}(1) - 0 \right] dy$$

$$= \int_0^a \left(\frac{a^2-y^2}{2} \right) \frac{\pi}{2} dy = \frac{\pi}{4} \int_0^a (a^2-y^2) dy$$

$$= \frac{\pi}{4} \left[a^2 y - \frac{y^3}{3} \right]_0^a = \frac{\pi}{4} \left[a^3 - \frac{a^3}{3} \right]$$

$$= \frac{\pi}{4} \frac{2a^3}{3} = \underline{\underline{\frac{\pi a^3}{6}}}$$

$$\int_1^2 \int_{-\sqrt{2-y}}^{\sqrt{2-y}} 2x^2 y^2 dx dy$$

$$= \int_1^2 2y^2 \left(\frac{x^3}{3} \right)_{-\sqrt{2-y}}^{\sqrt{2-y}} dy \quad \left| \int (ax+b)^n = \frac{(ax+b)^{n+1}}{(n+1)a} \right.$$

$$= \int_1^2 2y^2 \left[\frac{(2-y)^{3/2}}{3} + \frac{(2-y)^{3/2}}{3} \right] dy$$

$$= 2 \int_1^2 y^2 \frac{2}{3} (2-y)^{3/2} dy$$

$$= \frac{4}{3} \int_1^2 y^2 (2-y)^{3/2} dy$$

$$\text{put } 2-y = t$$

$$y = 2-t$$

$$dy = -dt$$

$$y: 1 \rightarrow 2$$

$$t: 1 \rightarrow 0$$

$$= \frac{4}{3} \int_1^0 (2-t)^2 t^{3/2} (-dt)$$

$$= \frac{4}{3} \int_0^1 (4-4t+t^2) t^{3/2} dt$$

$$= \frac{4}{3} \int_0^1 (4t^{3/2} - 4t^{5/2} + t^{7/2}) dt$$

$$= \frac{4}{3} \left[\frac{4t^{5/2}}{5/2} - \frac{4t^{7/2}}{7/2} + \frac{t^{9/2}}{9/2} \right]_0^1$$

$$= \frac{4}{3} \left[\frac{4 \times 2}{5} - \frac{8}{7} + \frac{2}{9} \right] = \frac{4}{3} \left[\frac{8}{5} - \frac{8}{7} + \frac{2}{9} \right]$$

$$= \frac{856}{945}$$

$$\int_0^{\pi/4} \left[\int_{r=0}^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr \right] d\theta$$

put $1+r^2 = t$

$$2rdr = dt \Rightarrow rdr = \frac{dt}{2}$$

$$r: 0 \rightarrow \sqrt{\cos 2\theta} \quad t: 1 \rightarrow 1 + \cos 2\theta$$

$$= \int_0^{\pi/4} \int_1^{1+\cos 2\theta} \frac{dt}{2t^2} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \int_1^{1+\cos 2\theta} t^{-2} dt d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[\frac{t^{-1}}{-1} \right]_1^{1+\cos 2\theta} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[-\frac{1}{t} \right]_1^{1+\cos 2\theta} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[-\frac{1}{1+\cos 2\theta} + 1 \right] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[-\frac{1}{2\cos^2\theta} + 1 \right] d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left[-\frac{1}{2} \sec^2\theta + 1 \right] d\theta$$

$$= \frac{1}{2} \left[-\frac{1}{2} \tan\theta + \theta \right]_0^{\pi/4} = \frac{1}{2} \left[-\frac{1}{2} (1) + \frac{\pi}{4} - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$\int_0^{\pi/2} \int_0^{a \cos \theta} r \sqrt{a^2 - r^2} dr d\theta$$

$$a^2 - r^2 = t \quad \Rightarrow \quad -2r dr = dt$$

$$r dr = -\frac{dt}{2}$$

$$r: 0 \rightarrow a \cos \theta$$

$$t: a^2 \rightarrow a^2 - a^2 \cos^2 \theta$$

$$a^2 \sin^2 \theta$$

$$= \int_0^{\pi/2} \int_{a^2}^{a^2 \sin^2 \theta} \sqrt{t} \left(-\frac{dt}{2} \right) d\theta$$

$$\left| (t^{1/2}) \right.$$

$$= -\frac{1}{2} \int_0^{\pi/2} \left(\frac{t^{3/2}}{3/2} \right)_{a^2}^{a^2 \sin^2 \theta} d\theta$$

$$= -\frac{1}{2} \int_0^{\pi/2} \left((a^2 \sin^2 \theta)^{3/2} - (a^2)^{3/2} \right) d\theta$$

$$= -\frac{1}{3} a^3 \int_0^{\pi/2} (\sin^3 \theta - 1) d\theta$$

$$\left| \int_0^{\pi/2} \sin^3 \theta \cos^0 \theta d\theta \right.$$

$$= -\frac{a^3}{3} \left[\int_0^{\pi/2} \sin^3 \theta d\theta - [\theta]_0^{\pi/2} \right] = \frac{1}{2} B\left(\frac{1}{2}, \frac{3}{2}\right)$$

$$= -\frac{a^3}{3} \left[\frac{1}{2} B\left(2, \frac{1}{2}\right) - \frac{\pi}{2} \right] = -\frac{a^3}{3} \left[\frac{1}{2} \frac{\Gamma(2) \Gamma(1/2)}{\Gamma(5/2)} - \frac{\pi}{2} \right]$$

$$= -\frac{a^3}{3} \left[\frac{1}{2} \frac{\sqrt{\pi}}{3/2 \cdot 1/2 \sqrt{1/2}} - \frac{\pi}{2} \right] = -\frac{a^3}{3} \left[\frac{2}{3} - \frac{\pi}{2} \right]$$

$$\int_0^{\pi/2} \int_0^{1-\sin\theta} r^2 \cos\theta \, dr \, d\theta$$